Mutian (Austin) He

San Jose, CA | 408-548-1060 | mhe3@scu.edu | LinkedIn | Github | Portfolio

EDUCATION

Santa Clara University, Leavey School of Business **Jan 2025 - Jan 2027**
Master of Science, Information Systems *Santa Clara, US*
- **Coursework:** Cloud Computing, Database Systems, Machine Learning, Deep Learning, Reinforcement Learning, NLP

University of Glasgow, School of Computing Science **Sep 2021 - Jun 2024**
Bachelor of Science, Computer Science *Glasgow, UK*
- **Coursework:** Data Structures & Algorithms, Database Systems, Software Engineering, Human-Computer Interaction

EXPERIENCE

Amazon Web Services (AWS) **Jan 2026 - Jun 2026**
AI Engineer Intern Live Demo

Project Description: Built an AI-driven platform called RiskLens AI that transforms unstructured SEC 10-K filings into structured risk insights, enabling cross-year risk comparison and decision support. Processed 348 filings across 76 companies and 11 industries.

- Designed and implemented an end-to-end data pipeline on AWS to parse, clean, and structure financial filings from SEC EDGAR, using S3 for hierarchical storage (industry/company layout) and Bedrock for downstream LLM consumption.
- Built a ReAct-based multi-step LLM agent using AWS Bedrock Converse API with 6 custom tools (risk loading, cross-company comparison, keyword search, stock quotes, news aggregation), enabling autonomous multi-turn risk analysis.
- Developed a dual-model architecture (DeepSeek V3 for dialogue orchestration + Amazon Nova Pro for structured extraction), with three-dimensional risk scoring (financial impact, likelihood, urgency) validated by deterministic Python logic.
- Implemented a hybrid classification framework combining rule-based keyword mapping with LLM fallback for low-confidence cases, achieving consistent 9-category risk taxonomy across 11,500+ risk factors.
- Built a full-stack web application (React + Vite frontend, Python FastAPI backend) featuring interactive dashboard with risk heatmap, cross-company comparison, financial tables extraction via AWS Textract, and real-time stock/news integration.

PROJECTS

Goal-Conditioned Maze Navigation with Curriculum Learning | Github **Apr 2026 - May 2026**

Project Description: Designed and implemented a goal-conditioned RL environment in MuJoCo PointMaze, training a PPO agent to navigate configurable mazes with curriculum learning and reward shaping.

- Implemented PPO with goal-conditioned observation space in MuJoCo PointMaze, achieving 38% success rate with curriculum learning vs. 6% baseline on complex mazes.
- Designed a three-stage curriculum (UMaze → Medium → Large) with automatic promotion based on rolling success rate, enabling progressive policy transfer across increasing maze complexity.
- Conducted systematic ablation studies comparing sparse vs. dense reward shaping and curriculum vs. direct training, with trajectory visualization and reward curve analysis.

Image Search System | Live Demo | Github **Mar 2026 - Apr 2026**

Project Description: Built a content-based image retrieval system that enables visual similarity search over large-scale image datasets, using deep learning feature extraction and efficient vector indexing.

- Extracted high-dimensional image embeddings using a pre-trained ResNet-50 model and indexed them with FAISS for approximate nearest-neighbor search, achieving sub-second retrieval on 100K+ images.
- Designed a Flask-based web interface with drag-and-drop upload, real-time similarity ranking, and configurable distance metrics (cosine, L2).
- Implemented PCA-based dimensionality reduction to compress feature vectors from 2048-d to 256-d, reducing index size by 87% while maintaining top-10 retrieval accuracy above 95%.

SKILLS

- **Programming:** Python, SQL (MySQL, PostgreSQL), Java Script, HTML/CSS, MongoDB
- **Machine Learning & AI:** PyTorch, Reinforcement Learning (PPO, Policy Gradient fundamentals), Model Evaluation, Feedback-driven Optimization, Multimodal Learning, Representation Learning, LLM Agents, Prompt Engineering
- **Simulation & RL Tools:** Gymnasium, Stable-Baselines3 (SB3), MuJoCo / Isaac Gym, Reward Design, Training & Evaluation Loops
- **Data Quality & Evaluation:** Output Evaluation, Feedback Signal Design, Human-in-the-loop Validation, Data Quality Assessment
- **Data Engineering:** Apache Kafka, Apache Airflow, ETL/ELT Pipelines, Streaming Architecture, Data Modeling
- **Cloud & Platforms:** AWS (EC2, S3, Glue, Athena, Lambda), Docker, Railway, Cloudflare